

Supervised (labeled Data) Machine learning

- model tries to create best func that maps input to output (features to target)
- takes both x and y as inputs
- tells on this x , there will be this y .
- the change of this x will now affect the y .

Regression (predict) (predictive model) Classification (put them in a group) (descriptive model)

1
2.1 {goal is to predict
3 a continuous numerical
4 value for an input)
4000
500

continuous predicting house price
forecasting sales

1
2 {goal is to predict a category
1 or class label}.
2
1
1

categorical multi class classif
is (0-9 number)
is more than 2 val

• Handwritten digit
recognition
• Email spam detection
(True/False - 0/1 - Yes/No.)
Binary class classification

- both will have target.

UnSupervised ML (unlabeled Data) → output

- Model tries to understand pattern in the feature to create different groups/clusters on the basis of similarity.

- takes x as input.

user tells how many groups we want.
then the model breaks them into them.

- look into insights and group/cluster

- good cluster → similarities with own type
differences with diff type



How is ML different from Simple AI Agents

1. Beyond Rule-Based System.

- ↳ doesn't fail on new examples/input
- ↳ learn on the new data and make it its own.
- ↳ pattern learn and give answer accordingly.

2. Machine learning Agent (similar problem) (bcz abduction have limits, (will outsmart abductive))

- ↳ allows agents to learn from experience
- ↳ behavioral pattern, likes, dislikes → personalized ads.
- ↳ recommendation system, self driving cars

3. ML makes agent smarter (learning)

↳ reinforcement learning

↳ points out uncertainty, error and improve the error in next learning batch.

Machine Learning (K-FOLD CROSS VALIDATION)

• A way of computers to learn patterns from data.

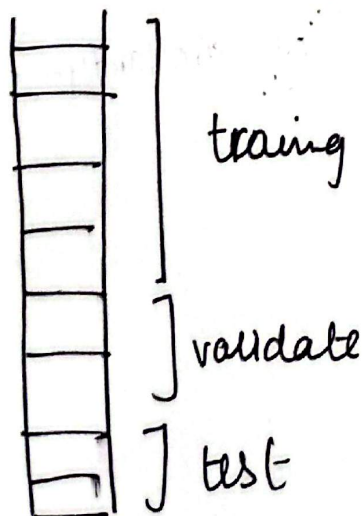
1. Predictive Analytics (csv)

2. Computer vision (image)

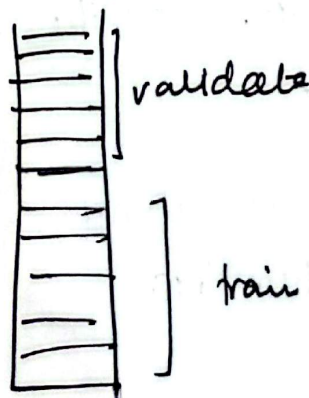
3. Natural language processing (chatbots)

* test set will be used after training

* validation set will be used during training.



1st iter



2nd iter

